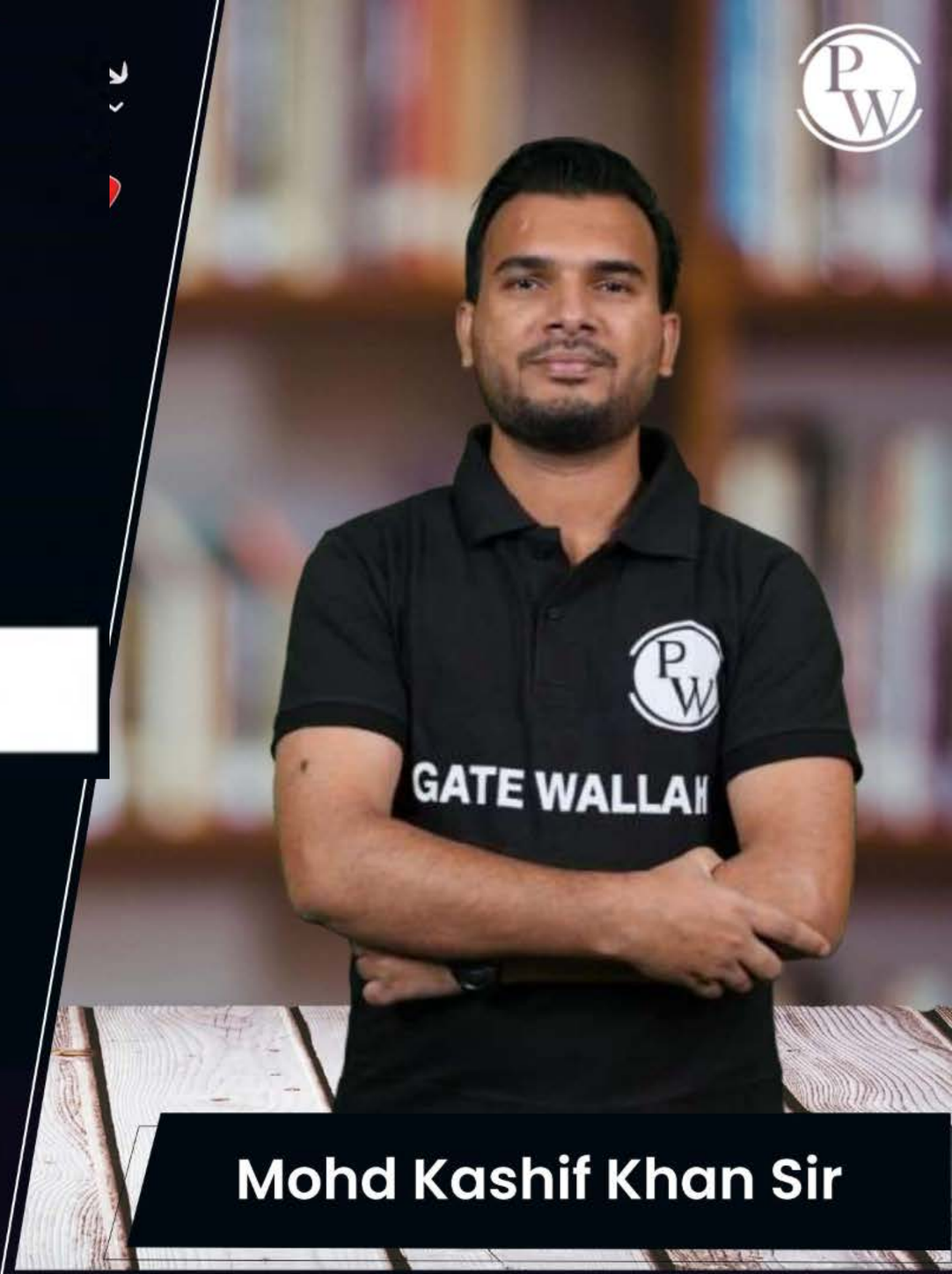
COMPUTER SCIENCE & DA

Data Structure through Python

Lists and Arrays and Stack

DPP- 01

Discussion Notes



Mohd Kashif Khan Sir

[MCQ]

#Q What is the postfix representation of the following infix expression?
(A + B) * C - D * E / F

A AB + C * DE * F - /

B A B * C + D E * F / -

C AB + C - DE * F/‘

D AB + C * DE * F/-



AB + C * DE * F / -

[MCQ]

#Q What is the outcome of the prefix expression +, – , * , 3, 2, /, 8, 4, 1 ?

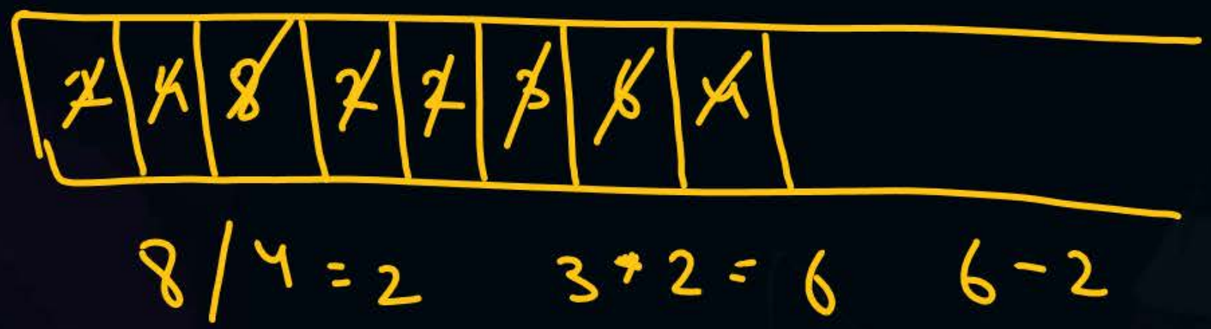


A 12

B 5

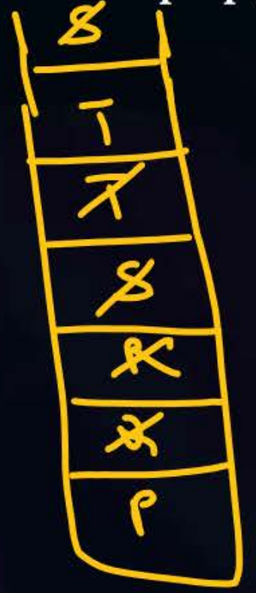
C 11

D 4



[MCQ]

#Q The five items P, Q, R, S and T are pushed in a stack, one after the other starting from P. The stack is popped four times and each element is inserted in a queue Then two elements are deleted from the queue and pushed back on the stack, now one item is popped from the stack. The popped item is:





A P

B R

C Q

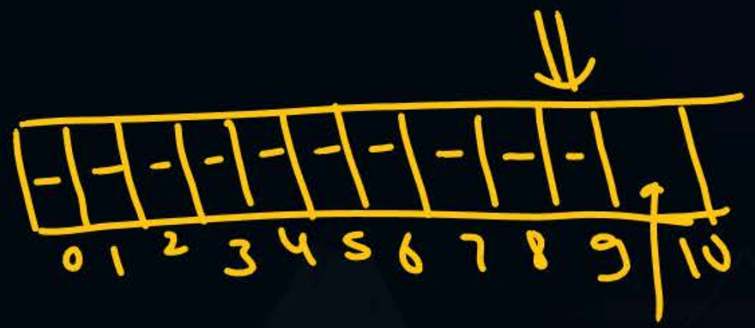
D S

[NAT]

#Q Consider the following Stack implement using the stack

```
    SIZE = 11
    class Stack:
        def __init__(self):
            self.arr = [0] * SIZE # Initialize array with zeros
    self.top = -1 # Initialize top pointer
# Example usage:
stack = Stack()
```

What would be the maximum value of the top that does not cause the overflow of the stack?



[MCQ]

#Q A stack is normally used in digital computers to store the return address at the time of a call because

funA( )
L1
L2 ⇝ funB( )
⇒ L3
funA( )

funB( )
L1
L2 funC( )
→ L3

funC( )
L
L2
L3

A Stacks are non-volatile tile memories

B Stacks have large capacity

C Information in a stack cannot be altered by other instructions

D Stacks permit easy nesting of subroutine



[NAT]

#Q Consider the following sequence of operations on an empty stack. push(54); push(52); pop(); push(55); push(62); s = pop();
Consider the following sequence of operations on an empty queue. enqueue{21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q = dequeue(); The value of s + q is

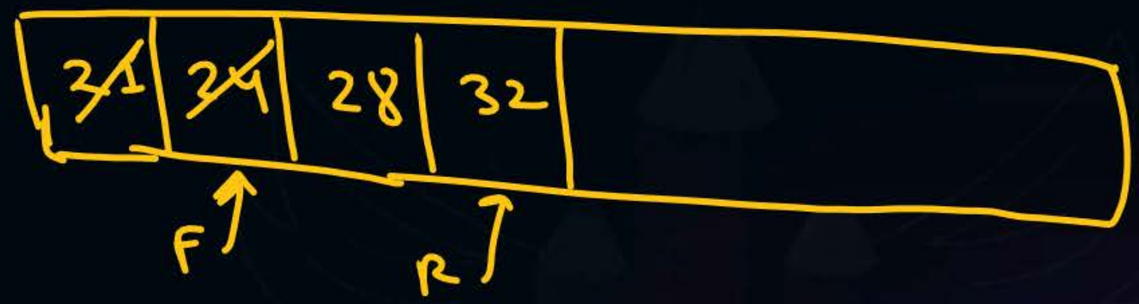


s = 62

s + q = 62 + 24

= 86

q = 24



THANK - YOU